



A Simple Guide to Database Design

As the title says, this is an introduction to databases, their purpose and features. It deals with how to design and create a simple database in MySQL. Starting off with the concepts of data and information, it is also a good read for those new to databases.

It's a Saturday morning, as I feed these words into a word processor. I have a browser running in the background, with a page on eBay open in one tab, and Engadget in another. The sheer simplicity of browsing through the product catalogues on eBay, or reading the latest and greatest tech news on Engadget, is living proof of the power of the database. The information age, the social networking age, the collaborative age, or whatever 'age' you may want to call it, all revolve around data.

Shouldn't the information age be revolving around, well, information? At this point, I would like to highlight a key difference between information and data. Information is processed data. From a philosophical standpoint, it has been wonderfully put by Gene Bellinger, Durval Castro and Anthony Mills in their article, aptly titled, 'Data, Information, Knowledge, and Wisdom': the cumulative order would be Data < Information < Knowledge < Understanding < Wisdom.

As far as computing goes in day-to-day contexts, the buck stops at information. Databases, which, as the word suggests, deal with what underlies information, i.e., data. Consider the product page on eBay. All you see is information, the cohesive form, which consists of the product images, the description, the price, information about the seller, etc. The data beneath this is processed and presentation by the browser. In this example, the individual components are data—like the words in the description. The words, by themselves, can't stand independently on the page. Only when they are in context with one another, and with other data,

do they convey something that makes sense. When they are processed, they turn into information.

The domain of databases is vast, and fills volumes and volumes of mammoth publications. The scope of this article is to give you a taste and understanding of databases, how to go about designing one, and I will conclude with designing an actual but hypothetical database, and implementing it.

So, what are databases?

Databases are structures that hold data. The software that enables the flow of data through these structures is called a database management system or DBMS. The most widely used system of DBMS is something called RDBMS or Relational DBMS. This simply means that the data is stored in tables; moreover, whatever relationships that exist within the data are stored within tables.

To put it simply, there are three parts that make a database:

1. **Tables:** We all know what tables are—a matrix of rows and columns. In databases, it's the same. Each row is a record, or a unit of data. A record (row) can have several columns or fields. Each field is like an attribute of that record.
2. **Queries:** A query is a question posed to the database, to retrieve a specific set of records, based on conditions supplied in the query.
3. **Views:** These are virtual tables, or (a set of) stored queries.